

**INTELLIGENT ACADEMIC ADMINISTRATION & STUDY
CANDIDACY MANAGEMENT MODULE FOR URESEARCH 2.0
CGS DASHBOARD**

by

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Interim Report submitted in partial fulfilment of
the requirements for the
Bachelor of Information Systems (Hons)

Final Year Project I
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CERTIFICATION OF APPROVAL

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Information Systems
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BACHELOR OF INFORMATION SYSTEMS (HONS)

Approved by,



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CERTIFICATION OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is my own except as specified in the references and acknowledgements, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.



CHLOE CHING QING EN

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ABSTRACT

The Centre for Graduate Studies (CGS) at Universiti Teknologi PETRONAS currently manages several postgraduate administrative processes manually, including but not limited to workstation allocation, study candidacy reminders, study candidacy appeals, and the dismissal of students who have exceeded their study candidacy. These processes depend heavily on manual checking, email communication, document verification, and manual database updates, which leads to repetitive work, longer processing times, and a higher risk of human error. This project proposes an Intelligent Academic Administration and Study Candidacy Management Module as part of the UResearch 2.0 CGS Dashboard, an initiative that integrates six administrative modules into one centralised platform. The proposed module digitalises the four existing CGS workflows without changing any university policies or business rules. It introduces workflow automation with sequential approval routing, automated email notifications, an operational dashboard for monitoring, and an AI Academic Guidance Assistant that answers student enquiries based on CGS standard operating procedures and university regulations. The AI assistant provides guidance only and does not make any approval decisions. Requirement gathering has been completed through semi-structured interviews with CGS administrative staff and analysis of existing workflow documents and forms. The project adopts the Iterative Waterfall model, as the business processes are stable and validated, while limited iterations allow refinements after stakeholder feedback. The planned system will be developed as a web application using HTML, CSS, JavaScript, Bootstrap, and PHP with a MySQL database, supported by Power Automate for workflow automation and Microsoft Copilot Studio for the AI assistant. At this stage, the system design and architecture have been established, and development will be carried out in FYP II, followed by testing, user acceptance testing, and evaluation.

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CHAPTER 1

INTRODUCTION

1.1 Background of Study

UResearch is an existing research management platform used by Universiti Teknologi PETRONAS (UTP) to support postgraduate academic administration and research-related processes. The platform initially acts as a repository for postgraduate academic information without any advanced features. It is currently being enhanced into UResearch 2.0, which aims to include the Centre for Graduate Studies (CGS) module and improve operational efficiency by integrating multiple CGS administrative modules into a single, centralised dashboard. The enhanced version, UResearch 2.0, consists of six integrated modules that will be developed by different students, and this project is responsible for one of the modules, which is the ‘Intelligent Academic Administration and Study Candidacy Management Module’.

The Centre for Graduate Studies (CGS) is the department responsible for handling and managing the postgraduate academic administration at UTP. At present, many of the day-to-day administrative workflows are still performed manually. For example, workstation layout is prepared using PowerPoint slides, and study candidacy status is checked one by one before reminder emails are sent by the CGS staff. While these methods can get the tasks done, they are entirely manual. Thus, they are making CGS staffs prone to human error and requiring them to perform repetitive and complex tasks which significantly reducing efficiency.

This project proposes the development of an Intelligent Academic Administration and Study Candidacy Management Module as part of the UResearch 2.0 CGS Dashboard to provide a solution that addresses the issues stated. Requirement gathering conducted with CGS administrative staff confirmed that this project does not redesign the

university's business processes and does not change any existing policies or business rules.

The aim of this project is to digitalise and automate the mentioned manual workflows which include the following business processes: Workstation Management, Study Candidacy Reminder, Study Candidacy Appeal, and Dismiss Exceed Study Candidacy. In addition to automating these workflows, the module includes an operational dashboard for monitoring, automated email notifications, and an AI Academic Guidance Assistant that helps students understand policies, procedures, and required documents.

1.2 Problem Statement

The existing CGS administrative workflows rely heavily on manual operations, which creates four main problems.

- **Inefficient Manual Administrative Processes:** Graduate administration involves multiple processes such as study candidacy monitoring, appeals, workstation allocation, and administrative approvals. CGS staff explained that most processes currently rely on manual checking, email communication, document verification, and database updates, which results in repetitive administrative work and a higher risk of human error.
- **Fragmented Workflow Management:** Graduate administrative activities often involve multiple stakeholders, including students, supervisors, Chair of department, Dean of Postgraduate and Research (PGR), and multiple Centre of Graduate Studies (CGS) staff. Different CGS staff are responsible for different business processes and manage the records independently. Furthermore, appeal process often passes through multiple approval stages involving different stakeholders. Without a centralized dashboard, it is difficult to track application progress, coordinate approvals, and maintain consistent records.
- **Limited Process Monitoring and Notification Support:** The existing UResearch platform only act as a repository instead of process management.

As a result, important milestones such as study candidacy expiry, appeal deadlines, and workstation allocation require manual monitoring by administrative staff without real time updates. The lack of automated reminders and centralized monitoring lead to higher risk of missed deadlines, delayed actions, and increased administrative burden.

- **Limited Guidance for Administrative Procedures:** Postgraduate students are currently obtain information regarding administrative procedures, policy, and guideline by manually searching the supporting documents or inquiring with CGS staff directly. It places the workload on CGS staff and also increases the students' waiting time.

1.3 Objectives

The objectives of this project are as follows:

- To digitalise and automate the manual administrative workflows of the Centre for Graduate Studies (CGS) that covers proposed business processes through sequential approval routing, automated deadline checking, email reminders, and automatic database updates.
- To develop a centralised operational dashboard that allows CGS staff and approvers to monitor application progress, approval status, and candidacy records within a single, centralised platform.
- To provide an AI Academic Guidance Assistant that answers student enquiries on administrative procedures, policies, and required documents based on existing CGS documentation without involving the AI in any decision-making.

1.4 Scope of Study

As the proposed module forms part of the larger UResearch 2.0 initiative, it is necessary to clearly define what this project covers and what remains outside its boundaries. The scope of study define by its target users, the business processes included, the functions to be developed, and area that are not included.

1.4.1 Included Processes

Four business processes are included in this module: Workstation Management, Study Candidacy Reminder, Study Candidacy Appeal, and Dismiss Exceed Study Candidacy. These processes were confirmed during requirement gathering with CGS staff.

1.4.2 Target Users

The module mainly serves three groups of users. The first group is postgraduate students, who submit applications and appeals, view workstation availability, and receive reminders and notifications. The second group is CGS administrative staff, who verify applications and monitor student candidacy status. The third group is the approvers, consisting of the Supervisor, the Chair of Department, and the Dean of Postgraduate and Research (PGR), who review and endorse on appeal applications.

1.4.3 In Scope

Within four proposed processes, the project also delivers several system functionalities: workflow automation, sequential approval routing, an operational dashboard embedded within the website, automated email notifications, and an AI Academic Guidance Assistant. The dashboard displays operational information only, such as pending appeals, reminder status, available and occupied workstations.

1.4.4 Out of Scope

Several areas are excluded to keep the project focused and achievable. The other five UResearch 2.0 modules are handled by other team members and are outside this project's boundary, although the system design is aligned so that all modules can integrate within the same dashboard. The Registry's internal dismissal process remains outside the project scope; the system only prepares the dismissal candidate list for CGS review and updates records after Registry confirmation. University policies and business rules will not be changed, and the AI assistant will not perform any decision making.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This chapter reviews existing studies on the digitalisation of academic administration in higher education and uses them to establish where the proposed module contributes. The reviews are map directly to the three objectives stated in Chapter 1. No published systems have been found that fully address the specific combination of problems in graduate processes mentioned in Chapter 1. Therefore, some of the evidence presented below comes from relevant fields.

2.2 Digitalisation of Academic Administration

Abishov et al. (2014) reported an automated information system for university management, arguing that efficiency gains appear only when an institution moves from disconnected subsystems to one covering administrative activity as a whole. Their implementation provided centralised storage, single-point document execution across departments and automated email alerts, and they add a caution this project takes seriously, that such a system is an instrument of management while the final decisions stay with the responsible officers. The paper is a design description rather than an evaluation, however, and reports no measured outcome.

Duan and Zhang (2007) designed a web-based management information system for graduate education. They decomposed it according to the service flows of graduate education rather than the organisational chart of the administrative department, and extended access beyond administrators to students and supervisors. After deployment for more than two thousand students they reported an average reduction of roughly 75% in work period, although the paper does not describe how this was measured. Both studies speak to the fragmentation described in Section 1.2, and both predate cloud

services and low-code platforms, so their architectural reasoning transfers to this project while their implementation path does not.

2.3 Workflow Automation and Sequential Approval Routing

The closest published work is Adedapo et al. (2023), who developed a workflow management system for a department in a Nigerian university. Their diagnosis matches the CGS situation, in that generic workflow tools are seldom tailored to universities and lack support for multi-stage academic approvals. Their system delivered automated notification, real-time status updates and configurable approvals, and users reported reduced delays. Those benefits are self-reported rather than measured against a baseline, so the study shows the approach is acceptable without quantifying the time saved. Their chains are also user-configurable, whereas the CGS appeal route is fixed by regulation from student to Supervisor, Programme Chair, CGS verification and finally the Dean of PGR, which makes a fixed route through one shared Approval Portal easier to build and to validate.

Bhardwaj and Kumar (2025) widen the picture through a review of 54 studies published between 2020 and 2024, finding that rule-based administrative tasks are now routinely automated using platform-based tools rather than purpose-built engines, and that cost, data security and change management are the principal barriers. Their headline figures need qualification, since several of the largest reported savings come from vendor customer stories rather than peer-reviewed evaluation. The direction of the evidence holds even so, and supports orchestrating the CGS routing and reminder scheduling in Power Automate rather than writing a scheduler and approval state machine in PHP, keeping that layer maintainable after handover. Neither study handles a recurring notification that must terminate when any of several unrelated conditions is met, the requirement described in Section 3.5.2.

2.4 Administration of Postgraduate Candidature

Mewburn et al. (2014) interviewed twenty research candidates about progress reporting, and their findings cut against the assumption that a formal reporting instrument produces transparency. Participants frequently did not know where a submitted report went, who read it, or what happened next, and the authors conclude

that management tools such as the progress report are not, by their nature, instruments that necessarily produce fair or equitable outcomes. The study is diagnostic and proposes no system, so its value here lies in establishing cost rather than solution. Administrative opacity generates anxiety independently of the eventual outcome, and the authors note non-completion rates of around one third in Australia and up to forty per cent in the United States, which places administrative delay in a more serious light than a clerical framing suggests.

2.5 Process Monitoring and Operational Dashboards

Schwendimann et al. (2017) reviewed 55 dashboard papers published between 2010 and 2015. Dashboards were found to support monitoring and improve awareness, but the reviewed systems overwhelmingly targeted teachers and students, and administrators appear as an intended user group in only a handful of the tabulated cases. Their review is confined to learning contexts, and this scarcity of administrative dashboards is itself the finding worth reporting.

Sarikaya et al. (2019) supply the design vocabulary Schwendimann et al. do not provide for this purpose. From a corpus of 83 dashboards spanning multiple domains they derived fifteen design factors grouped under purpose, audience, visual and interactive features, and data semantics. Their purpose dimension separates strategic, tactical and operational decision support, and their account of an operational dashboard fits the CGS case closely, in that it reports the current and recent past through immediately quantifiable metrics attributable to the entity responsible for them. They also argue that design principles do not transfer freely between dashboard types. Their framework must be used for what it is, since the authors state that they consulted no designers or users directly and that it is descriptive rather than prescriptive: it can establish that an operational dashboard is the correct category for CGS but cannot specify how that dashboard should be laid out.

Every indicator confirmed with CGS nonetheless answers what needs action today, each is attributable to a responsible party, and none calls for trend analysis, so an embedded operational dashboard is the correct category, while a separate business

intelligence layer would add overhead without serving the monitoring need identified in Section 1.2.

2.6 AI Assistance for Administrative Guidance

Katsamakas et al. (2024) map the effects of artificial intelligence on a university and single out administration and operations as an area attracting less attention than teaching and research. They recommend bounded assistants for students, staff and administrators as a practical entry point, list hallucination, accuracy and completeness as recurring weaknesses of generative tools, and conclude that AI should augment human roles rather than replace them. Their model is qualitative and the authors acknowledge that computational modelling is the next step, so the paper offers a direction for scoping an assistant rather than evidence that a particular design works.

Cao et al. (2023) deployed four role-separated chatbots in a computer science course, evaluating responses on accuracy, fluency, empathy, engagement and relevance. Their pilot indicates that a language model can produce accurate and human-like answers, and they stress that interaction quality governs acceptance. The setting is pedagogical, so what transfers is not the content but the structural finding that a bounded role behaves more predictably than a general-purpose assistant.

Oreški and Vlahek (2024) close most of the remaining distance. They grounded a student-support chatbot in faculty regulations and website content through a low-code retrieval pipeline covering administrative rules, required documents and appeal procedures. Their functional requirements include a detail this project adopts: a query falling outside the knowledge base returns the responsible officer's contact details and raises a ticket rather than an answer. Testing involved fifteen students in one faculty with no accuracy figures reported, so the evidence is preliminary, but the escalation pattern addresses in design the risk Katsamakas et al. (2024) identify in principle.

2.7 Summary of Reviewed Studies

Table 1 summarises the ten studies reviewed and the limitation of each.

Table 1: Summary of Reviewed Studies

Title	Author(s)	Year	Focus	Key Finding Relevant to This Project	Limitation
Transforming higher education with RPA	Bhardwaj & Kumar	2025	Review of 54 studies on automation in higher education, 2020–2024	Rule-based administrative tasks are automated using platform tools, not custom engines.	Largest reported savings come from vendor case studies, not peer-reviewed evaluation.
AI and the transformation of higher education institutions	Katsamakas, Pavlov & Saklad	2024	Causal loop model of AI transformation in a university	Administration is under-examined; bounded assistants are the practical entry point.	Qualitative model only; authors note computational modelling is still needed.
Retrieval augmented generation in LLMs	Oreški & Vlahek	Sept 2024	Chatbot grounded in faculty regulations	Escalates to a named officer when a query falls outside the knowledge base.	Tested with fifteen students; no accuracy figures reported.
Workflow management system for higher education	Adedapo et al.	Dec 2023	Workflow system for a university department	Automated notification and approval routing reduce administrative delay.	Chains are user-configurable; benefits self-reported, not measured.
AI chatbots as multi-role pedagogical agents	Cao et al.	Aug 2023	Four role-separated chatbots in a CS course	Role-bounded assistants behave more predictably than general-purpose ones.	Pedagogical, not administrative; preliminary pilot only.
What do we talk about when we talk about dashboards?	Sarikaya et al.	Jan 2019	Design space from 83 dashboards across several domains	Operational dashboards report current metrics attributable to a responsible party.	Descriptive not prescriptive; no designers or users consulted.
Perceiving learning at a glance	Schwendimann et al.	2017	Review of 55 learning dashboards, 2010–2015	Administrators are rarely the intended dashboard users.	Confined to learning contexts; low evaluation maturity.
Experiencing the progress report	Mewburn, Cuthbert & Tokareva	2014	Interviews with twenty doctoral candidates	Students often cannot see where their case sits or who acts on it.	Diagnostic study; no system.
Automated information system for university management	Abishov et al.	2014	Integrated system for university management	Integration across subsystems, not local automation, produces the gain.	No measured outcome; predates cloud platforms.
Web-based MIS for academic degree and graduate education	Duan & Zhang	2007	Web-based MIS for graduate education	Decomposition by service flow; access extended to students and supervisors.	Measurement method for the reported 75% reduction is not described.

2.8 Research Gaps

Based on the reviewed studies, four research gaps are identified.

Gap 1: Lack of an integrated graduate administration platform covers all processes that CGS handles. Most existing studies focus on specific administrative functions, such as graduate record management, progress reporting, or general university information systems. They do not provide a centralised platform that integrates multiple postgraduate administrative processes, such as workstation management, candidacy reminders, appeals, and dismissal, into a single system

Gap 2: Approval routing is either manual or generically configurable, never regulation-bound. Automated routing has been demonstrated, but with chains users can reconfigure. No reviewed system implements a route fixed by regulation with approve, return, reject and comment actions at each stage followed by rule-based calculation of the outcome, and none handles a recurring reminder that must stop when any of several unrelated conditions is met.

Gap 3: Insufficient intelligent academic guidance for students. Current AI chatbot research primarily focuses on teaching, learning engagement, and academic support. There are limited AI applications for postgraduate administration, insufficient guidance on candidature procedures and university policies, and limited AI support for administrative enquiries.

Gap 4: Limited monitoring and early intervention for study candidacy. Progress monitoring in the literature relies mainly on periodic reporting. There is limited continuous monitoring of study candidacy, a lack of automated reminder mechanisms, and limited early intervention for students who are approaching candidature expiry.

2.9 How the Proposed Project Addresses the Gaps

The proposed module responds to each of these gaps in a practical way. For Gap 1, the module integrates four postgraduate administrative processes into one centralised module within the UResearch 2.0 CGS Dashboard, which itself integrates six modules into a single platform. For Gap 2, the module implements workflow automation with sequential approval routing through an Approval Portal, automated email notifications, and automatic database updates after final approval are replacing the current email communication. For Gap 3, the module introduces an AI Academic Guidance Assistant built on CGS standard operating procedures, university regulations, administrative guidelines, existing forms, and frequently asked questions, providing administrative guidance rather than teaching support. For Gap 4, the module performs daily checks of candidacy status in the database and will send monthly reminders automatically beginning three months before expiry date. The reminders automatically stop when the conditions are met, and updates the overall reminder status on the operational dashboard to allow CGS staff intervene early where needed.

CHAPTER 3

METHODOLOGY AND PROJECT WORK

3.1 Development Methodology: Iterative Waterfall Model

This project adopts the Iterative Waterfall model as its development methodology. The Iterative Waterfall model follows the sequential phases as the traditional Waterfall approach, which moving from requirements to analysis, design, coding, testing, and maintenance. However, it adds backward feedback loops between phases. This allows the project to revisit and correct issues in each phase without waiting until the end of the project. Figure 1 shows the phases and feedback loops of the Iterative Waterfall model.

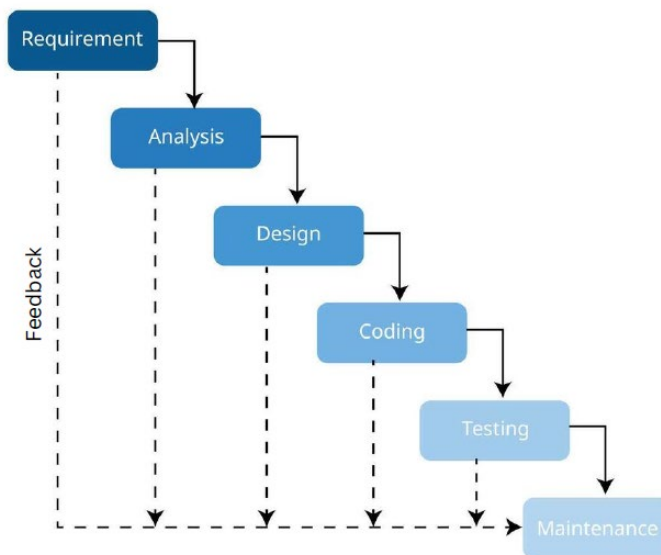


Figure 1: Iterative Waterfall Model

The Iterative Waterfall model is more suitable for this project than the traditional Waterfall model or Agile. Compared with the traditional Waterfall model, the key difference is the feedback loop. The traditional model assumes each phase is

completed perfectly before the next begins and emphasize the development phase should not be re-visited after it's done, which is unrealistic in practice; if a misunderstanding about a workflow is discovered during development, the traditional model provides no structured way to return to the design phase. The Iterative Waterfall model handles this by adding the feedback loop in each phases, allowing limited revisions based on stakeholders feedback, which fits this project's plan of validating each part of the system with CGS staff as work progresses.

3.2 Research Activities

The project work is organised into six phases, as summarised in Table 2.

Table 2: Project Phases and Activities

Phase	Activities
Requirement Gathering	Conduct interviews with CGS staff and collect business process requirements.
System Analysis	Analyse existing manual workflows and identify system requirements.
System Design	Design the system architecture, database, user interface, and workflow automation.
Module Development	Develop the proposed module and integrate Power Automate, MySQL, and the AI Guidance Assistant.
System Testing	Perform functional testing and User Acceptance Testing (UAT).
Documentation	Prepare technical documentation and the final dissertation.

Currently, requirements gathering is completed, and business processes have been confirmed with CGS staff. Based on the confirmed business processes, system analysis has been completed. The system design phase for integrated view of six modules is completed. System design for specific modules is ongoing. Development work has not yet started. Therefore, this report does not contain any implementation deliverables. Module development, system testing, and the remaining documentation such as technical paper and dissertation are expected to complete in FYP II.

3.3 Data Collection

Both primary and secondary data were collected to support the project.

3.3.1 Primary Data Collection Methods

The first primary data collection method was a semi-structured interview with CGS administrative staff. The purpose of the interview was to understand the current manual workflows, administrative procedures, and system requirements. A semi-structured format was chosen because it allowed prepared questions to guide the session while leaving chance for follow-up questions when staff described the details of each process.

The second method was document analysis of CGS administrative documents, including workflow charts, application forms, and related administrative documents such as the appeal memo, the study candidacy reminder template, and the exceed study candidacy due date template. The purpose was to analyse the existing business processes and the functional requirements for digitalisation, and to confirm exactly what data each process captures.

3.3.2 Secondary Data Collection Methods

Secondary data consisted of journal articles and conference papers, which were reviewed in Chapter 2 to identify research gaps and support the proposed system design. University policies and administrative guidelines is also considered as secondary data, which were studied to understand the study candidacy regulations and business rules. Examples include the Postgraduate Workstation Terms of Use, the postgraduate onboarding slides, and the postgraduate handbook. Existing UResearch documents were also referred to so that the module remains consistent with the current platform.

3.4 Requirement Gathering Findings

An interview was conducted with one CGS staff, Amirul Hariz Yunus, who responsible for postgraduate administration on 23 June 2026. The findings are summarised below.

Finding 1: Current administrative workflows rely heavily on manual work. Most processes depend on manual checking, email communication, document verification, and database updates. CGS staff described this work as repetitive and time-consuming, and it also creates opportunities for human error when records are updated by hand across separate files.

Finding 2: Existing business rules should remain unchanged. CGS confirmed that the university's policies, such as the maximum appeal duration of twelve months and the required approval sequence, must be preserved in the digitalised system.

Finding 3: The sequential approval workflow should be retained but automated. The appeal process must continue to move from the student to the Supervisor, the Programme Chair, CGS verification, and finally the Dean, but the routing between stages should be handled by the system.

Finding 4: Four processes were identified as priorities for digitalisation. When asked which processes require the most administrative effort, CGS staff identified the study candidacy reminder, appeal processing, dismissal management, and workstation allocation.

3.5 Business Process As-Is Vs To-Be

This section compares the four processes as they are currently performed and the proposed solution. Figure 2 compares the current and proposed processes for all four workflows.

3.5.1 Workstation Management

CGS currently prepares the workstation layout in PowerPoint, students apply through Microsoft Forms, and staff assign seats one by one, with locker keys handled through a separate application. Allocation and tracking are entirely manual, availability cannot be validated in real time, and keys are sometimes never collected because no follow-up exists. In the proposed process, availability is published on the system and students select a seat directly. The system validates the choice, updates the database, and sends a confirmation

email, or a rejection email where two students select the same seat simultaneously. CGS retains manual assignment for exceptional cases.

3.5.2 Study Candidacy Reminder

CGS staff currently check each student's status manually, prepare a reminder list, and send emails monthly from three months before expiry, giving three reminders in total. Students who have submitted softbound, become inactive, or completed their studies are removed from the list manually, and no record is kept of which reminders were sent. In the proposed process, candidacy status is checked daily and reminders are issued automatically on the same monthly schedule. Generation stops once an appeal is submitted, softbound is approved, or the student becomes inactive or dismissed, and reminder status is recorded and visible to CGS.

3.5.3 Study Candidacy Appeal

A student currently submits an appeal on an Excel template, which is endorsed by the Supervisor and the Programme Chair, verified by CGS, and approved by the Dean of PGR, with every stage coordinated by email. CGS then calculates the extension, updates the database, and sends the outcome manually. In the proposed process, the student submits a standardised online form and selects their Supervisor, and the system routes the appeal through the same four stages, where each approver may approve, reject, or return with comment. After the Dean approves, the extension is calculated, the database updated, the student notified, and the dashboard refreshed automatically. The business rule and university guideline now enforced by the system.

3.5.4 Dismiss Exceed Study Candidacy

Every month, CGS staff manually prepare a list of students who have exceeded their study candidacy, either because their candidacy due date has expired with no appeal or because they have already used the maximum twelve months of appeal. The list is submitted to the Registry for dismissal. After the Registry confirms, CGS manually updates the records and informs the students by email. This process is time-consuming, involves high manual work, and causes delays

in notification. In the proposed process, the system generates the candidate list from candidacy records for CGS to review and confirm. Submission to the Registry remains manual and the Registry's internal process stays outside the project scope, but once dismissal is confirmed the notification to the student is issued automatically.

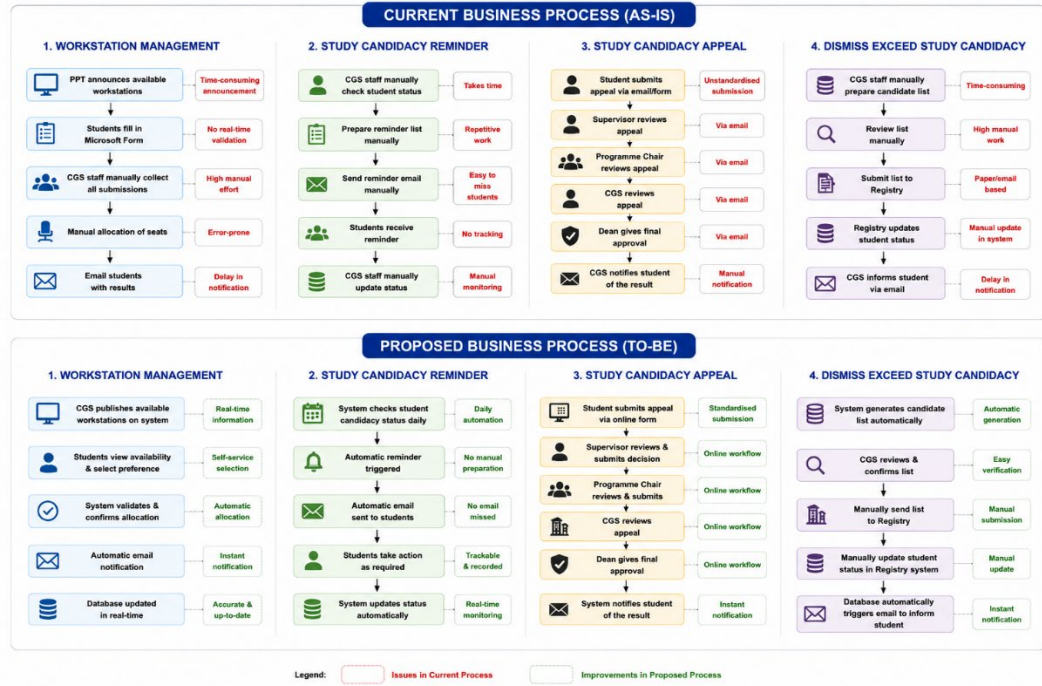


Figure 2: Business Process As-Is vs To-Be

3.6 System Architecture

The system architecture is organised into five layers under the UResearch 2.0 CGS Dashboard, with authentication handled through the existing UResearch login using single sign-on. Figure 3 shows the overall integrated system architecture and figure 4 shows the system architecture of the specific module that this project covers.

1. Presentation Layer: the web interface, consisting of the Student Portal, the Approver Portal, the CGS Staff Portal, and the Administrator Portal.
2. Application Layer: the core module, containing Workstation Management, Study Candidacy Reminder, Study Candidacy Appeal, Dismiss Exceed Study

Candidacy, the Operational Dashboard, the Notification Centre, and the AI Academic Guidance Assistant.

3. Workflow Layer: process automation using Microsoft Power Automate or n8n, together with PHPMailer where needed, handling sequential approval routing, reminder automation, workstation allocation automation, notification triggers, and database update automation.
4. Integration Layer: the AI and notification services, where Microsoft Copilot Studio provides the AI guidance capabilities and Microsoft Outlook handles automated email notifications for submissions, approvals and rejections, reminders, and status updates.
5. Data Layer: a MySQL database storing users and roles, student records, workstation records, appeal records, reminder schedules, approval statuses, and notification logs.



Figure 3: Integrated System Architecture

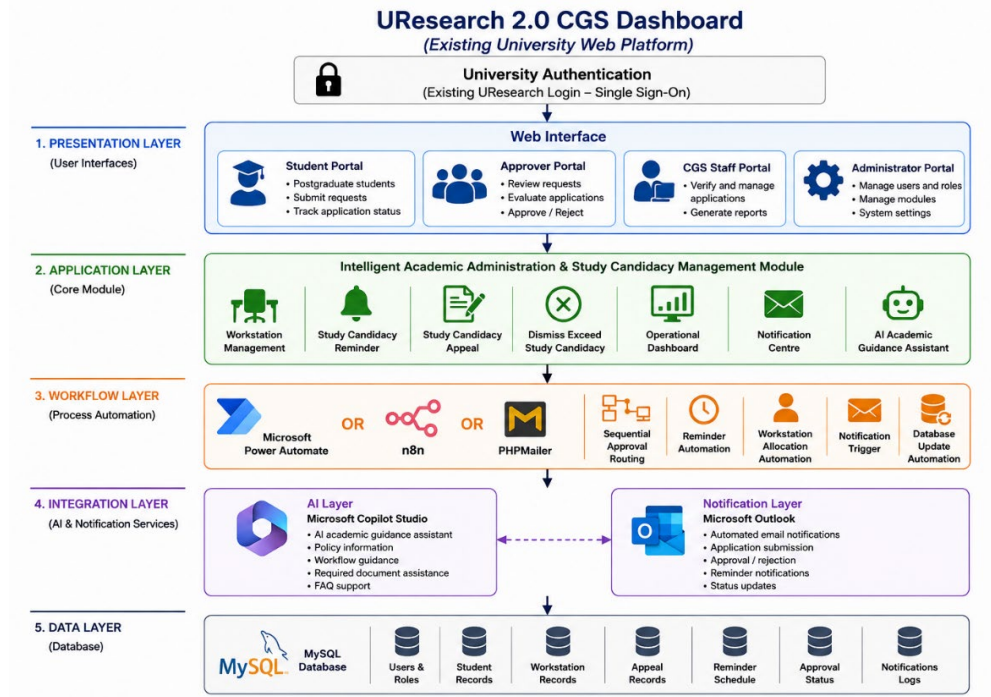


Figure 4: System Architecture of the Proposed Module

3.7 Proposed User Interface Design

The user interface follows a two-level structure. The first level is the main page, which is shared by all six UResearch 2.0 modules and was designed together with the other project members so that the platform presents one consistent environment. Each user group is directed to its own landing view according to the role held at login, so the main page is different in content and available actions between students, approvers, CGS staff and system administrators, while keeping the same layout and visual theme throughout. From the main page, the user selects the module required, and the second level opens the view belonging to that module.

This section presents the login page and four shared main pages. They are the first draft of designs that will be used in the implementation phase, minor modification will be applied during development phase in FYP II based on each module needs. They define the structure and theme that the screens of this project module should follow. The module-specific screens, covering workstation selection, candidacy appeal submission, reminder tracking and the operational dashboard, are reached from these

main pages and will be designed and developed during FYP II within the boundary set out in Section 1.4.

3.7.1 Login Page

Figure 5 shows the login page. A single entry point is provided for all user groups, with the role selected at login and access rights applied afterwards according to the role stored in the database. This keeps authentication consistent with the existing UTP system single sign-on described in Section 3.6.

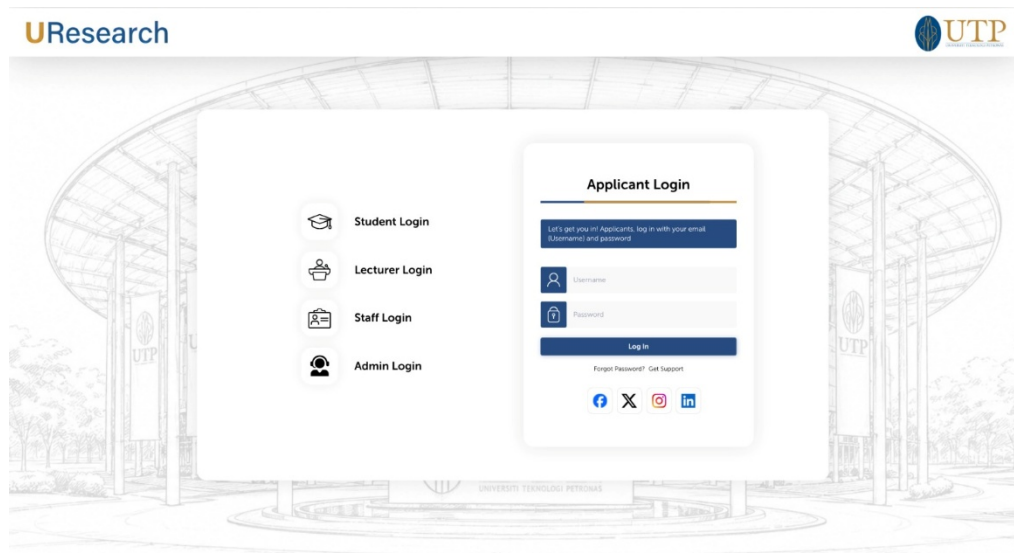


Figure 5: Login Page

3.7.2 Student Portal

Figure 6 shows the student landing page. The upper section summarises the student's own applications, showing how many are in progress and how many notifications are unread. Below this, the application list displays each submission with its current status, so a student can see how far a request has moved without contacting CGS. A notification panel and an upcoming task list occupy the lower half of the page. This layout addresses the visibility problem described in Section 1.2.

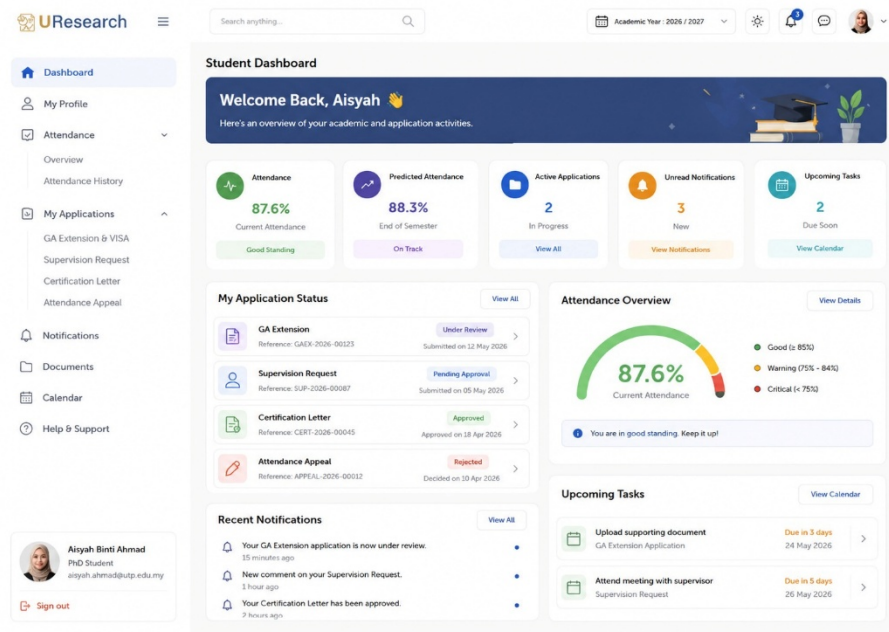


Figure 6: Landing Page – Student View

3.7.3 Approver Portal

Figure 7 shows the approver view, which is shared by the Supervisor, the Programme Chair and the Dean of PGR. Rather than providing a separate interface for each role, one portal displays only the applications currently waiting on that approver, since all three perform the same actions at different points in the same sequence. Pending, overdue and completed items are separated so that an approver can identify what requires action first. The approve, reject and return-with-comment actions described in Section 3.5.3 will be implemented within this portal.

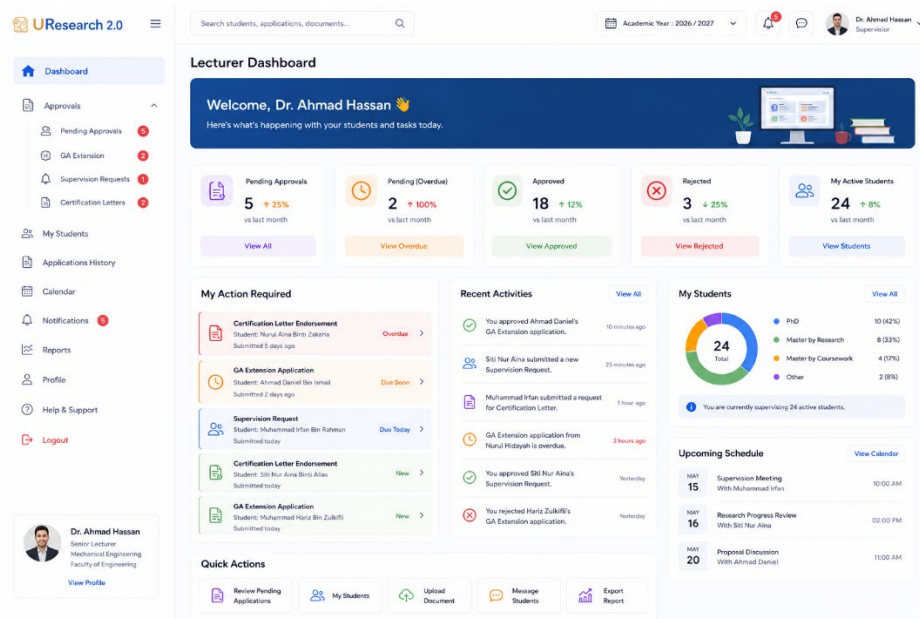


Figure 7: Landing Page – Approver View

3.7.4 CGS Staff Portal

Figure 8 shows the CGS staff view, which serves as the operational dashboard for administrative monitoring. The layout groups total applications, items pending staff action, and approved and rejected counts along the top, with a workload breakdown and a pending action list below. This is the view that will display the indicators confirmed during requirement gathering, namely pending appeals, reminder status, workstation occupancy and pending locker key collection. The design deliberately reports current status rather than historical trends, consistent with the operational rather than analytical role discussed in Section 2.5.

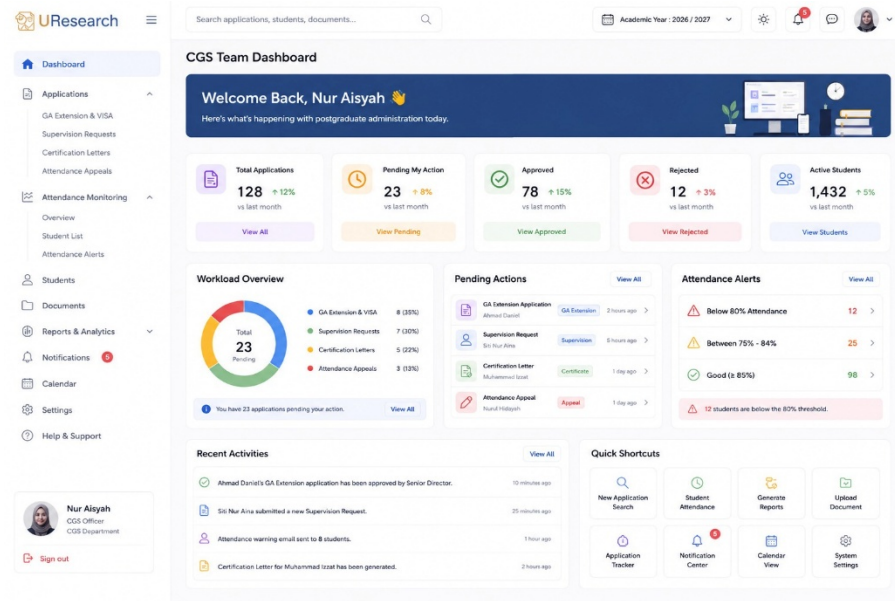


Figure 8: Landing Page – CGS Team View

3.7.5 System Administrator View

Figure 9 shows the system administrator landing page, which covers user and role management, system settings and audit logs. These functions belong to the UResearch 2.0 platform rather than to any single module. The page is included here to show how this module fits within the wider dashboard, and the administrator is not one of the three user groups defined in Section 1.4.2.

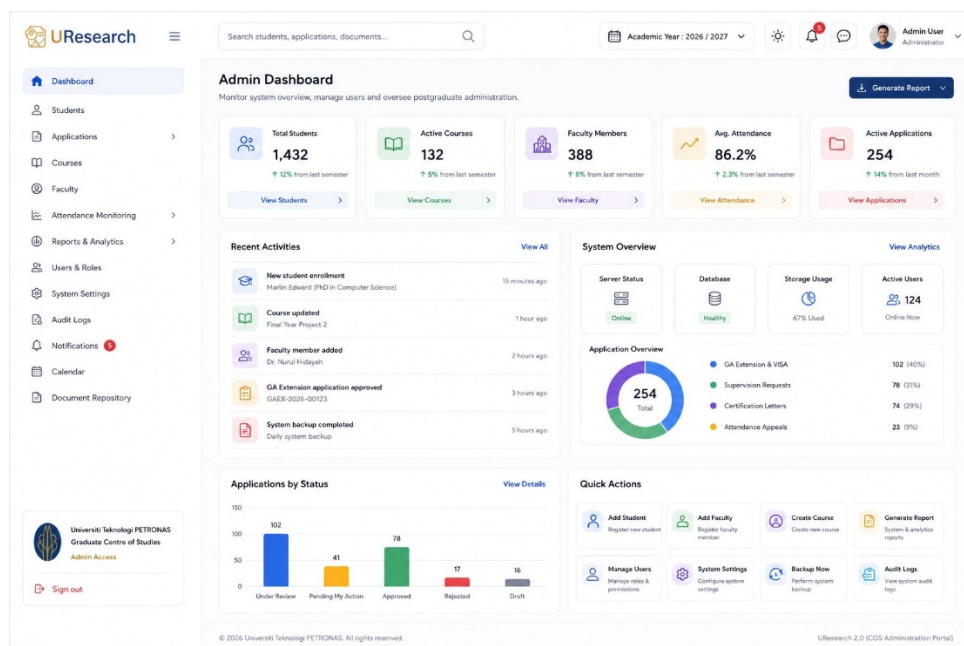


Figure 9: Landing Page – System Administrator View

3.8 Technology Stack

The technology stack keeps the implementation practical and consistent with a low-complexity, workflow-oriented approach. Table 3 lists the selected technologies. Alternatives are listed because licensing and integration constraints can only be confirmed during the development phase in FYP II.

Table 3: Technology Stack

Component	Technology
Frontend	HTML5
Styling	CSS3
UI Framework	Bootstrap 5
Client-side Logic	JavaScript
Charts	Chart.js
Backend	PHP 8
Database	MySQL
Database Tool	phpMyAdmin
Local Server	XAMPP
Workflow Automation	Power Automate or n8n
AI Assistant	Microsoft Copilot Studio or OpenAI API
Email	PHPMailer (SMTP) or Microsoft Outlook (Power Automate)
PDF Generation	Dompdf
Authentication	PHP Sessions with Password Hashing
Version Control	Git and GitHub
UI Design	Figma and Canva
UML and Diagrams	Draw.io and StarUML

3.9 Project Timeline

Figure 10 shows the project timeline for FYP I and FYP II.

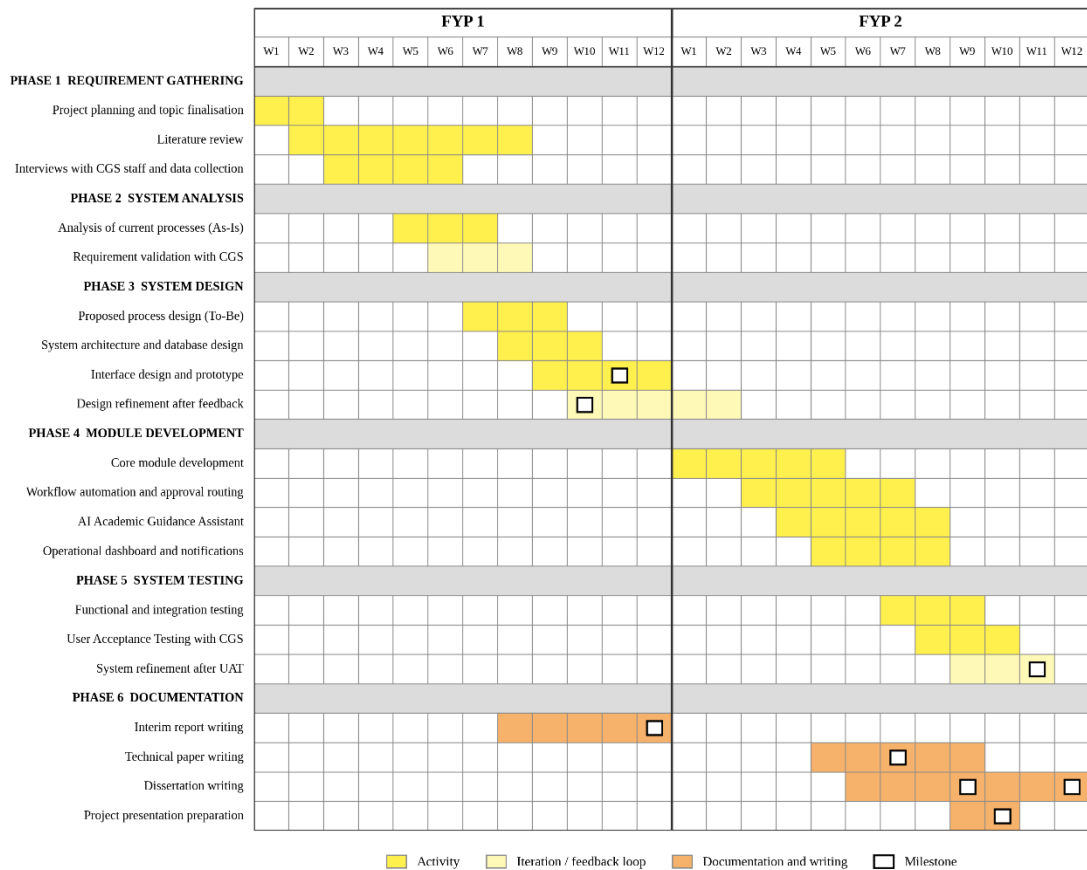


Figure 10: Project Timeline

CHAPTER 4

CONCLUSION AND FUTURE WORK

4.1 Conclusion

This project set out to digitalise four manual administrative workflows within the Centre for Graduate Studies: Workstation Management, Study Candidacy Reminder, Study Candidacy Appeal, and Dismiss Exceed Study Candidacy. Requirement gathering with CGS staff confirmed that these workflows depend on manual checking, email coordination and manual record updates, and that the existing business rules must be preserved while the repetitive steps are automated. Working from those validated requirements, FYP I has defined the project scope, adopted the Iterative Waterfall model, designed the four to-be processes, and established a five-layer architecture built on a PHP and MySQL web application with Power Automate and Microsoft Copilot Studio. The literature review found that no existing system combines all issues mentioned by CGS interviewee, which is the gap this project addresses. The analysis and design work are complete, and the project is ready to proceed to development.

4.2 Future Work

FYP II follows directly from this design. Development will begin with the core web application and MySQL database, followed by the sequential approval routing, automated reminder checks, workstation allocation logic and notifications, and then the AI Academic Guidance Assistant and operational dashboard. System testing will verify each workflow against the confirmed business rules, and User Acceptance Testing with CGS staff will assess whether the digitalised processes reduce manual administrative work as intended. The module will then be prepared for deployment within the UResearch 2.0 CGS Dashboard, alongside the technical paper and final dissertation.

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APPENDICES

Appendix A: Interview Questions

Section A: Respondent Background

- Q1. Could you briefly describe your role and responsibilities within CGS?
- Q2. Which postgraduate administrative processes are you directly involved in?

Section B: Overview of Current CGS Administrative Processes

- Q3. What are the major academic administration processes currently managed by CGS?
- Q4. Which processes generate the highest number of requests from students?
- Q5. Which processes require the most administrative effort or coordination?

Section C: Detailed Workflow Discovery

- Q6. Can you explain the workflow of this process from beginning to end?
- Q7. Who are the stakeholders involved in this process?
- Q8. What approvals are required throughout the process?
- Q9. Are there any decision points where applications may be approved, rejected, or returned?
- Q10. What happens after an application is approved?
- Q11. What happens if an application is incomplete or contains errors?

Section D: Forms, Documents and Data Requirements

- Q12. What forms are currently used for this process?
- Q13. What supporting documents are students required to submit?
- Q14. What information is typically captured in the application?
- Q15. Are there any policies, guidelines, or SOPs governing this process?

Section E: Pain Points and Challenges

- Q16. What are the most common problems encountered during this process?
- Q17. Which stages of the process take the longest time?
- Q18. Are there any repetitive manual tasks currently performed by CGS staff?

Q19. How do students currently track their application status?

Section F: Opportunities for Digitalisation

Q20. Which parts of the process do you believe could be automated or improved?

Q21. What information would you like to monitor through a dashboard?

Q22. What notifications or reminders would be useful for staff and students?

Section G: Process Prioritisation for Project Scope

Q23. If only 3–4 processes could be digitalised first, which would you prioritise?

Q24. Which processes would provide the greatest benefit if digitalised?

Q25. Are there any processes that should NOT be included due to complexity or low priority?

Section H: Future Requirements for UResearch 2.0

Q26. What key features would you expect from an improved digital workflow system?

Q27. Are there any existing UResearch features that should be reused or integrated?

Q28. Is there anything else you believe should be considered when designing this module?

Appendix B: Meeting Photos



17 May 2026

Group Discussion – Proposal Planning

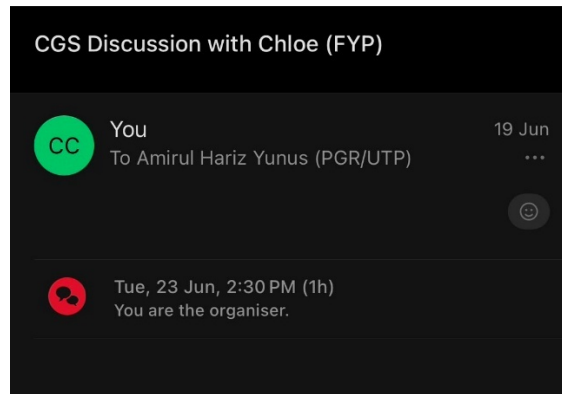
Finalized the project direction and proposal planning.



10 June 2026

First Meeting with CGS Staff

Gained an overview of the six UResearch 2.0 modules, current administrative workflows.



23 June 2026

Requirement Gathering with CGS Staff (Amirul Hariz Yunus)

Understood the existing workflows for the assigned processes and identified opportunities for workflow automation.



29 June 2026

Team Discussion – Module Integration

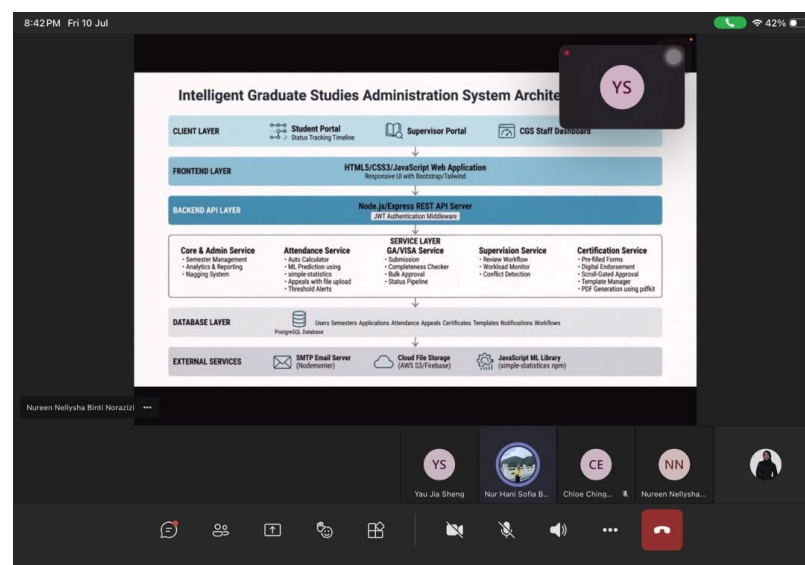
Discussed each member's module scope and aligned the system design to ensure integration within the UResearch 2.0 dashboard.



30 June 2026

Progress Meeting with Supervisor

Presented the current project progress and received guidance on the next development phase .



10 July 2026

System Design Discussion

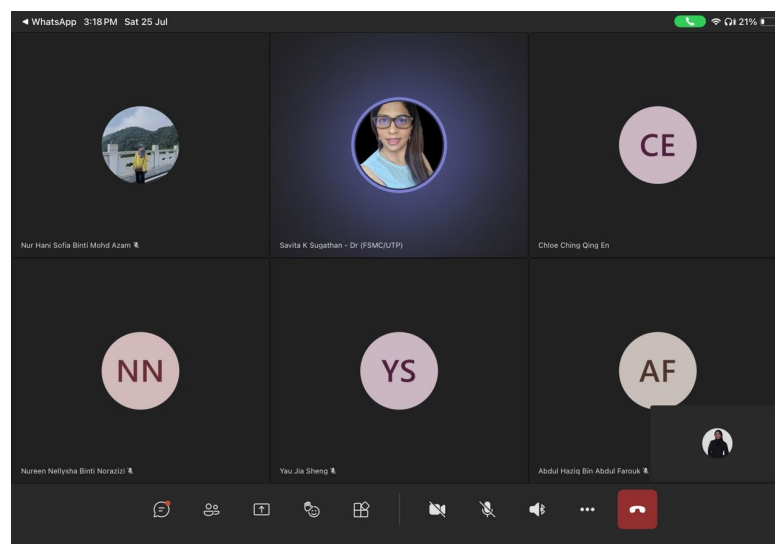
Discussed the proposed system architecture, technology stack, and low-fidelity prototype for the shared dashboard interface.



14 July 2026

Proposal Defence Preparation with Supervisor

Reviewed project progress and clarified the proposal defence presentation requirements and deliverables.



25 July 2026

Post-Proposal Defence Discussion with Supervisor

Discussed the examiner's feedback from the proposal defence and sought the supervisor's advice on the required improvements and revisions for the project.

Appendix C: Current Workflow Documents

 Attachment - Appeal for Extension Candidacy.xlsx 13 KB	 Memo - Appeal for Extension Candidacy.docx 49 KB	 Template of Study Candidacy Reminder.xlsx 24 KB	 Template - Exceed Study Candidacy Due Date.xlsx 25 KB
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Proposed Processes	Application forms	Supporting documents required from students	Policies, guidelines, or SOPs	Data/Information captured in the application
Workstation Management	https://forms.pharespoint.com/Forms/Default.aspx?Page=Application%20for%20GSC%20Postgraduate%20Workstation.aspx	Metric ID Picture (Front & Back)	Student Submitted > Supervisor Endorsement- Workstation & Drawer Assignment by GSC	As in the form
Study Candidacy Reminder	No Application Form	No Supporting Document	Reminder sent to student: • 3 months before due date (3 Times) • No soft-bound for visa submitted • No Withdrawal or Graduation record	In Attachment
Study Candidacy Appeal	In Attachment	In Attachment	• Appeal must be endorsed by Supervisor and Chair of department and approved by Dean of PGR • Maximum duration allowed by student to appeal is 12 months	• Student ID • Student Name • Programme • Supervisor Details • Chair of Department Details • Justification • Current Academic Progress
Dismiss Exceed Study Candidacy	No Application Form	In Attachment	Student Dismissed due to: • Exceed 12 months appeal to Dean of PGR • Expired Study Candidacy Due Date with no appeal	as in attachment

Documents Gathering through Email

	A	B	C	D	E	F	G	H	I	J	
1	A. Academic Progress/Current Status (Phase) *Please tick one										
2		1.Writing	Percentage of completion (%)								
3		2.Simulation									
4		3.Experiment									
5		4.Data Analysis									
6											
7		5. Research Completion Seminar (RCS) :				Date:					
8						Category:					
9											
10		or									
11		Expected date for RCS *if applicable				Date:					
12											
13		B. Candidacy Extension Record *Please tick one									
14		Through GSC	Yes								
15			No								
16		Through Vice Chancellor	Yes								
17			No								
18											
19		C. List of Publications/Journal									
20											
21		No	Description								
22											
23											
24											
25											
26											
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40											
41											
42											
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44											
45											
46			D. Duration of Extension Requested								
47											
48											
49			DISCLAIMER: All of the information provided by the applicant on this document is considered valid and accurate at the time it has been received. Any deceit or untruth may lead to discard of the application.								
50											
51											
52											

Attachment - Appeal for Extension Candidacy



MEMORANDUM

To : Professor Ts Dr Mohd Fadzil bin Hassan
Dean, Postgraduate & Research
Universiti Teknologi PETRONAS

Through : *HOD/Chair

Through : *Supervisor

From : *Student Name
*Department

Date :

Appeal for Extension of Candidacy

Dear [Professor](#) Ts Dr Mohd Fadzil bin Hassan,

Yours sincerely,

.....
*Student Name
*Student ID
*Department

Approved/ Not Approved

.....
Professor Ts Dr Mohd Fadzil bin Hassan
Dean
Postgraduate and Research

Memo - Appeal for Extension Candidacy

UTP Nexus Stay Connected Home Academic AppCentre Alumni Relations Office Student Affairs HSSE Innovation & Ventures Shop UTP Merchandise

Discover Publish Build OneDrive

CGS CENTRE FOR GRADUATE STUDIES

POSTGRADUATE WORKSTATION BLOCK N2

N2-02-01/02 (MALE)	N2-02-03/04 (MALE)
N2-02-05/06 (FEMALE)	N2-02-07/08 (FEMALE)
N2-02-10/11 (FEMALE)	N2-02-12/13 (MALE)
N2-02-14/15 (MALE)	

Power Point – Workstation Availability

Student Details

1. Student Name *

Enter your answer

2. Student Matric ID *

Enter your answer

3. Sponsorship *

Select your answer

4. Department *

Select your answer

Application for Workstation Agreement

Appendix D: Turnitin Result

Interim Report v1.0

ORIGINALITY REPORT

7 %
SIMILARITY INDEX

6 %
INTERNET SOURCES

4 %
PUBLICATIONS

3 %
STUDENT PAPERS

PRIMARY SOURCES

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2	utpedia.utp.edu.my Internet Source	1 %
3	etd.hu.edu.et Internet Source	<1 %
4	www.tandfonline.com Internet Source	<1 %
5	discol.umk.edu.my Internet Source	<1 %
6	internationaljournalcorner.com Internet Source	<1 %
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